

$$\textcircled{1} \quad a = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad b = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

$$v = 2a + b = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$$

$$|v| = 10$$

$$\hat{v} = \frac{v}{|v|} = \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}$$

$$v_1 = \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 2 \\ 2\sqrt{3} \end{pmatrix}, \quad \hat{v}_1 \cdot \hat{v}_2$$

$$|v_1| = \sqrt{(\sqrt{3})^2 + (1)^2} = \sqrt{4} = \underline{\underline{2}}$$

$$|v_2| = \sqrt{(2)^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = 4$$

$$\hat{v}_1 = \left(\frac{1}{2}\right) \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix}, \quad \hat{v}_2 = \left(\frac{1}{4}\right) \begin{pmatrix} 2 \\ 2\sqrt{3} \end{pmatrix}$$

$$\hat{v}_1 \cdot \hat{v}_2 = \left(\frac{1}{2} \times \frac{1}{4}\right) \left(\sqrt{3} \times 2 + 1 \times \underline{2\sqrt{3}}\right) = \frac{4\sqrt{3}}{8} = \sqrt{3}/2$$

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad |v_1| = \sqrt{1+1+4} = \sqrt{6}$$

$$|v_2| = \sqrt{4+1+1} = \sqrt{6}$$

$$v_1 \cdot v_2 = 1 \times 2 + 1 \times (-1) + 2 \times 1 = 2 - 1 + 2 = 3$$

$$v_1 \cdot v_2 = |v_1| \cdot |v_2| \cdot \cos(\theta) = 6 \cdot \cos \theta = 3$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

$$\underline{v_1} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \underline{v_2} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \underline{v_3} = \begin{pmatrix} 7 \\ 13 \end{pmatrix}$$

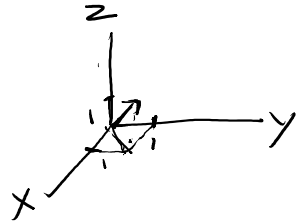
$$\underline{v_3} = \alpha \underline{v_1} + \beta \underline{v_2} \Rightarrow \alpha \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 13 \end{pmatrix}$$

$$\begin{pmatrix} 2\alpha \\ 2\alpha \end{pmatrix} + \begin{pmatrix} \beta \\ 3\beta \end{pmatrix} = \begin{pmatrix} 7 \\ 13 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2\alpha + \beta = 7 \\ 2\alpha + 3\beta = 13 \end{cases} \quad \begin{matrix} 2\beta = 6 \\ \underline{\beta = 3}, \quad \underline{\alpha = 2} \end{matrix}$$

$$\begin{pmatrix} 7 \\ 13 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 2 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \underline{v_3 = 2v_1 + 3v_2}$$

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$



$$v_2 = \alpha v_1 + \beta v_3$$

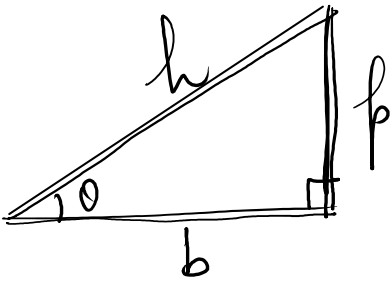
$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} \alpha + \beta \\ \alpha + 2\beta \\ \alpha + 2\beta \end{pmatrix}$$

$\alpha + \beta = 1$

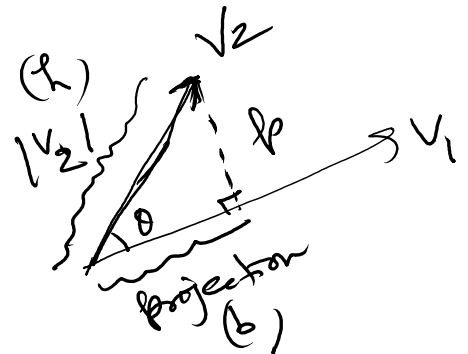
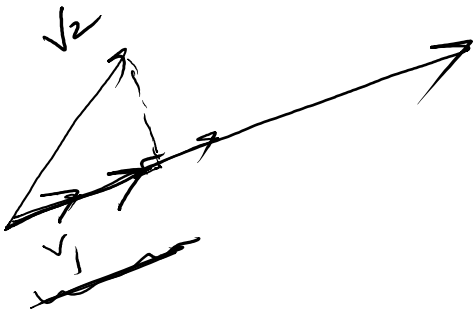
$\alpha + 2\beta = 2$

$\alpha + 2\beta = 3$



$$\begin{aligned}\cos\theta &= b/h \\ \sin\theta &= p/h \\ \tan\theta &= \frac{\sin\theta}{\cos\theta} = p/b\end{aligned}$$

$$\begin{aligned}b &= h \cos\theta \\ p &= h \sin\theta\end{aligned}$$



$$\text{projection} = |v_2| \cos\theta$$

$$= |v_2| \cdot \frac{v_1 \cdot v_2}{|v_1| |v_2|}$$

$$= v_2 \cdot \left(\frac{v_1}{|v_1|} \right) = \boxed{v_2 \cdot \hat{v}_1}$$

$$v_1 \cdot v_2 = |v_1| |v_2| \cos\theta$$

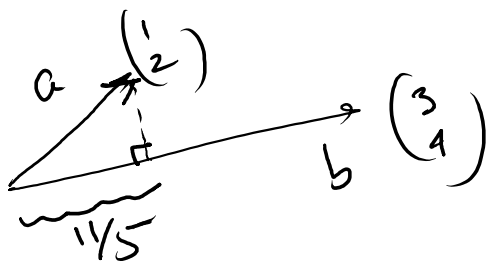
$$\Rightarrow \cos\theta = \frac{v_1 \cdot v_2}{|v_1| |v_2|}$$

$$a = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad b = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\text{projection of } a \text{ on } b = a \cdot \hat{b}$$

$$|b| = 5, \quad \hat{b} = \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}$$

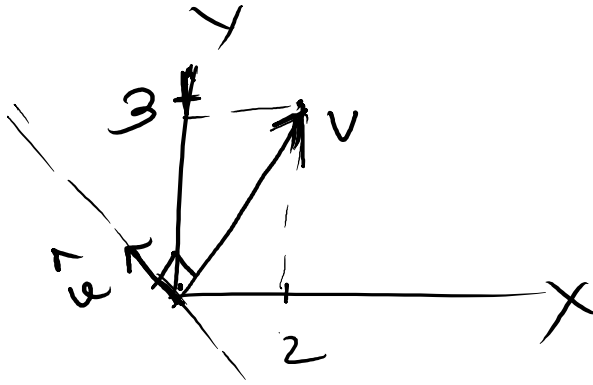
$$\text{projection} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix} = 3/5 + 8/5 = \underline{\underline{11/5}}$$



$$|a| = \sqrt{1+4} = \sqrt{5}$$



$$V = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$



$\vec{u}$

$$\vec{u} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \rightarrow \alpha^2 + \beta^2 = 1$$

$$V \cdot \vec{u} = 0$$

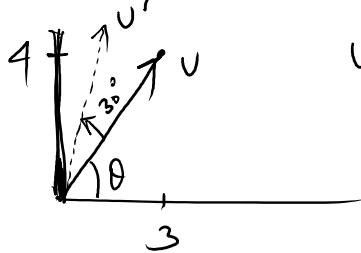
$$2\alpha + 3\beta = 0$$

$$\alpha = -3, \beta = 2$$

$$\sqrt{9+4} = \sqrt{13}$$

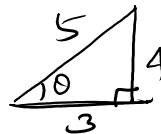
$$\vec{u} = \frac{1}{\sqrt{13}} \begin{pmatrix} -3 \\ 2 \end{pmatrix}$$

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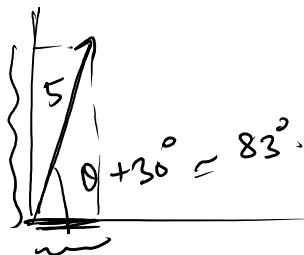
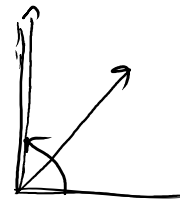
$$u = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$|u| = 5$$



$$\tan \theta = \frac{4}{3}, \quad \cos \theta = \frac{3}{5}$$

$\theta = ?$



$$\pi \text{ rad} = 180 \text{ deg.}$$

$$u' = \begin{pmatrix} 5 \cos(\theta + 30^\circ) \\ 5 \sin(\theta + 30^\circ) \end{pmatrix} = \begin{pmatrix} 0.60 \\ 4.96 \end{pmatrix}$$